Chapter 4: Conditional and Multivariate Distributions

Part I — Statistics Foundation for ML

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Abstract

Most interesting questions are about several quantities at once, and about what one tells you regarding another. This chapter sets up joint distributions, pulls out the marginal and conditional distributions that live inside them, and defines independence. Two identities — the law of total expectation and the law of total variance — let you compute a mean or a variance by conditioning, and the bivariate and multivariate normal give the formula for the conditional that reappears later as linear regression and as the Gaussian-process posterior.

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1 Joint, marginal, and conditional

For two variables X and Y the **joint** distribution is a mass function $f(x,y) = P(X=x, Y=y)$ (discrete) or a density $f(x,y)$ with $P\big((X,Y)\in A\big) = \iint_A f$ (continuous). Everything about the pair is in f; the single-variable distributions are recovered by summing or integrating the other variable away.

Definition 1.1 (Marginal and conditional)**.** The **marginal** of X is $f_X(x) = \sum_y f(x,y)$ (or $\int f(x,y)\,dy$). The **conditional** of Y given $X=x$, for $f_X(x) > 0$, is

$$f_{Y\mid X}(y \mid x) = \frac{f(x,y)}{f_X(x)}.$$

The conditional is itself a distribution in y (it is non-negative and sums to 1), so it has a mean of its own, the **conditional expectation** $E[Y \mid X=x] = \sum_y y\, f_{Y\mid X}(y \mid x)$. X and Y are **independent** when the joint factors, $f(x,y) = f_X(x)\, f_Y(y)$, equivalently when $f_{Y\mid X}(y \mid x) = f_Y(y)$ — knowing X tells you nothing about Y.

Everything in this section is mechanical once you have the joint table, which makes it a good thing to compute rather than memorise:

```
joint = {(0, 0): 0.3, (0, 1): 0.2, (1, 0): 0.1, (1, 1): 0.4}    # P(X=x, Y=y)

def marginal_x(j):
    out = {}
    for (x, y), p in j.items():
        out[x] = out.get(x, 0.0) + p
    return out
```

```python
def conditional_y_given_x(j, x0):
    px = sum(p for (x, y), p in j.items() if x == x0)
    return {y: p / px for (x, y), p in j.items() if x == x0}

print(marginal_x(joint))                 # {0: 0.5, 1: 0.5}
print(conditional_y_given_x(joint, 0))   # {0: 0.6, 1: 0.4}
print(conditional_y_given_x(joint, 1))   # {0: 0.2, 1: 0.8}  -- X shifts Y
```

Because $P(Y{=}1 \mid X)$ moves from 0.4 to 0.8 as X goes from 0 to 1, these two are *not* independent — X carries real information about Y.

2 Computing by conditioning

Conditioning is not just descriptive; it is a tool for computing means and variances of a single variable by splitting on another. Both laws below are used constantly, from mixture models to the bias–variance decomposition.

Theorem 2.1 (Law of total expectation)**.** $E[X] = E\big[E[X \mid Y]\big]$.

Proof. Write the inner expectation out and swap the order of summation:

$$\begin{aligned} E\big[E[X \mid Y]\big] &= \sum_y E[X \mid Y = y]\, f_Y(y) = \sum_y \sum_x x\, \frac{f(x,y)}{f_Y(y)}\, f_Y(y) \\ &= \sum_x x \sum_y f(x,y) = \sum_x x\, f_X(x) = E[X]. \end{aligned}$$

□

Theorem 2.2 (Law of total variance)**.** $\operatorname{Var}(X) = E\big[\operatorname{Var}(X \mid Y)\big] + \operatorname{Var}\big(E[X \mid Y]\big)$.

Proof. Let $m(Y) = E[X \mid Y]$. By definition $\operatorname{Var}(X \mid Y) = E[X^2 \mid Y] - m(Y)^2$, so $E[\operatorname{Var}(X \mid Y)] = E[X^2] - E[m(Y)^2]$ by the law of total expectation. Also $\operatorname{Var}(m(Y)) = E[m(Y)^2] - (E[m(Y)])^2 = E[m(Y)^2] - (E[X])^2$. Adding the two lines, the $E[m(Y)^2]$ terms cancel and $E[X^2] - (E[X])^2 = \operatorname{Var}(X)$ remains. □

In words: total variance is the average of the within-group variances plus the variance of the group means — "unexplained" plus "explained" spread. That split is exactly the bias–variance decomposition in disguise.

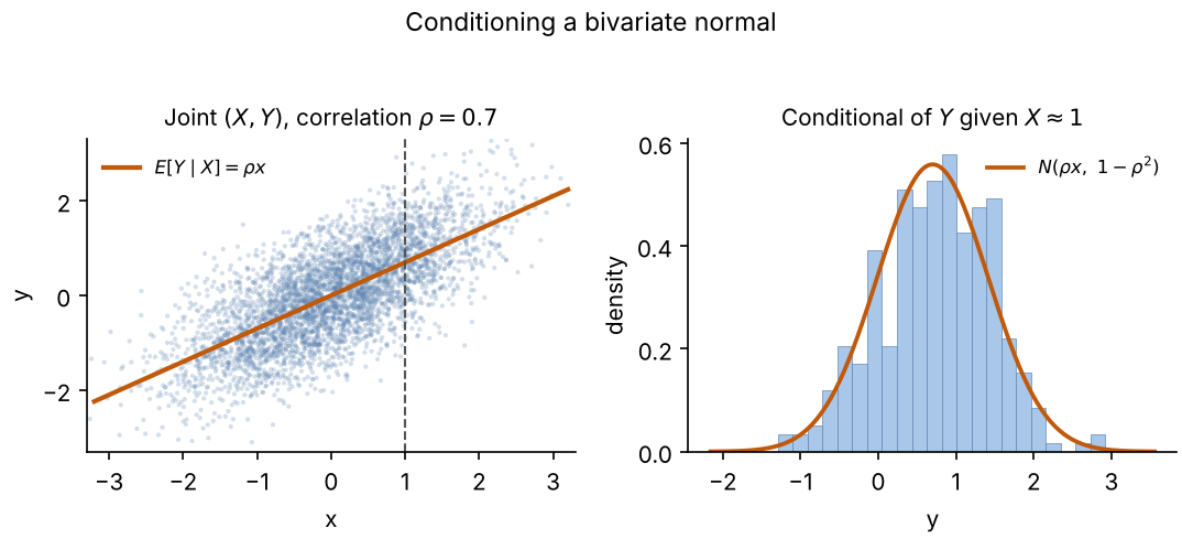
3 The bivariate and multivariate normal

For jointly normal variables the conditional distribution has a clean closed form, and it is linear in the conditioning value.

Theorem 3.1 (Bivariate normal conditional)**.** If (X, Y) is bivariate normal with means μ_X, μ_Y, standard deviations σ_X, σ_Y and correlation ρ, then $Y \mid X = x$ is normal with

$$E[Y \mid X = x] = \mu_Y + \rho\, \frac{\sigma_Y}{\sigma_X}\,(x - \mu_X), \qquad \operatorname{Var}(Y \mid X = x) = \sigma_Y^2\,(1 - \rho^2).$$

The conditional mean is a straight line in x — this is where linear regression comes from — and the conditional variance is *smaller* than the marginal σ_Y^2 by the factor $1 - \rho^2$: conditioning on a correlated variable removes a share of the uncertainty. The figure shows a sample with $\rho = 0.7$; the slice of Y at $X \approx 1$ matches the normal the formula predicts.



The same statement in full generality uses vectors and matrices. Partition a multivariate normal vector as (X_1, X_2) with mean (μ_1, μ_2) and covariance blocks $\Sigma_{11}, \Sigma_{12}, \Sigma_{21}, \Sigma_{22}$. Then $X_1 \mid X_2 = x_2$ is normal with

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2), \qquad \Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}.$$

This single formula is the engine of Gaussian-process regression in Part III: predicting at new inputs is conditioning a big multivariate normal on the observed outputs. A warning worth keeping: zero correlation does *not* mean independence in general — but for variables that are *jointly* normal, uncorrelated does imply independent, because $\rho = 0$ collapses the conditional mean to μ_Y and the joint density factors.

Summary

A joint distribution $f(x, y)$ contains the marginals f_X, f_Y (sum the other variable away) and the conditionals $f_{Y|X} = f(x, y)/f_X(x)$; independence is $f = f_X f_Y$. The law of total expectation ($E[X] = E[E[X \mid Y]]$) and the law of total variance ($\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E[X \mid Y])$) compute by conditioning. For the normal, $X_1 \mid X_2$ is again normal with a linear mean $\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$ and reduced covariance $\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$.